

ALFAAR40086

## Sodium fluoride, Puratronic, 99.995% (metals basis excluding K), K 60ppm approx.

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** Sodium fluoride, Puratronic, 99.995% (metals basis excluding K), K 60ppm approx.  
Sodium fluoride, Puratronic, 99.995% (metals basis excluding K), K 60ppm approx.

**Cat No. :** R40086  
**CAS No** 7681-49-4  
**Molecular Formula** F Na

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Powder Solid

**Appearance**  
White

**Odor**  
Odorless

#### Emergency Overview

Toxic if swallowed. Causes skin irritation. Causes serious eye irritation. Contact with acids liberates very toxic gas. Moisture sensitive.

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 3 |
| Skin Corrosion/Irritation         | Category 2 |
| Serious Eye Damage/Eye Irritation | Category 2 |

#### Label Elements



Sodium fluoride, Puratronic, 99.995% (metals basis excluding K), K 60ppm approx.

**Signal Word****Danger****Hazard Statements**

H301 - Toxic if swallowed

H315 - Causes skin irritation

H319 - Causes serious eye irritation

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P330 - Rinse mouth

P332 + P313 - If skin irritation occurs: Get medical advice/attention

P337 + P313 - If eye irritation persists: Get medical advice/attention

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Contact with acids liberates very toxic gas.

**Health Hazards**

Toxic if swallowed. Causes skin irritation. Causes serious eye irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component       | CAS No    | Weight % |
|-----------------|-----------|----------|
| Sodium fluoride | 7681-49-4 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

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**Most important symptoms and effects**

None reasonably foreseeable. Other side effects may include nausea, headache, diarrhea, cramps, vomiting, or flu-like symptoms

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Non-combustible. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from acids.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

## Sodium fluoride, Puratronic, 99.995% (metals basis excluding K), K 60ppm approx.

| Component       | China | Taiwan                     | Thailand                   | Hong Kong                  |
|-----------------|-------|----------------------------|----------------------------|----------------------------|
| Sodium fluoride | -     | TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> |

  

| Component       | ACGIH TLV                  | OSHA PEL                             | NIOSH                                                     | The United Kingdom                                                    | European Union |
|-----------------|----------------------------|--------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|----------------|
| Sodium fluoride | TWA: 2.5 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup> | IDLH: 250 mg/m <sup>3</sup><br>TWA: 2.5 mg/m <sup>3</sup> | STEL: 7.5 mg/m <sup>3</sup> 15 min<br>TWA: 2.5 mg/m <sup>3</sup> 8 hr |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

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**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                           |                                          |
|------------------------------------------------|---------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                     |                                          |
| <b>Physical State</b>                          | Powder Solid              |                                          |
| <b>Odor</b>                                    | Odorless                  |                                          |
| <b>Odor Threshold</b>                          | No data available         |                                          |
| <b>pH</b>                                      | 7.4                       |                                          |
| <b>Melting Point/Range</b>                     | 993 °C / 1819.4 °F        |                                          |
| <b>Softening Point</b>                         | No data available         |                                          |
| <b>Boiling Point/Range</b>                     | 1700 °C / 3092 °F         |                                          |
| <b>Flash Point</b>                             | No information available  | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable            | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available  |                                          |
| <b>Explosion Limits</b>                        | No data available         |                                          |
| <b>Vapor Pressure</b>                          | 1 mmHg @ 1077 °C          |                                          |
| <b>Vapor Density</b>                           | Not applicable            | Solid                                    |
| <b>Specific Gravity / Density</b>              | 2.78 (H <sub>2</sub> O=1) |                                          |
| <b>Bulk Density</b>                            | No data available         |                                          |
| <b>Water Solubility</b>                        | 40g/L (25°C)              |                                          |
| <b>Solubility in other solvents</b>            | No information available  |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                           |                                          |
| <b>Autoignition Temperature</b>                | No data available         |                                          |
| <b>Decomposition Temperature</b>               | No data available         |                                          |
| <b>Viscosity</b>                               | Not applicable            | Solid                                    |
| <b>Explosive Properties</b>                    | No information available  |                                          |
| <b>Oxidizing Properties</b>                    | No information available  |                                          |
| <b>Molecular Formula</b>                       | F Na                      |                                          |
| <b>Molecular Weight</b>                        | 41.98                     |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                        |
|---------------------------------|--------------------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions. Moisture sensitive.    |
| <b>Hazardous Reactions</b>      | None under normal processing.                          |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.               |
| <b>Conditions to Avoid</b>      | Incompatible products. Exposure to moist air or water. |
| <b>Materials to avoid</b>       | Acids.                                                 |

**Hazardous Decomposition Products** Gaseous hydrogen fluoride (HF).

## SECTION 11. TOXICOLOGICAL INFORMATION

### Product Information

#### (a) acute toxicity;

| Component       | LD50 Oral               | LD50 Dermal        | LC50 Inhalation |
|-----------------|-------------------------|--------------------|-----------------|
| Sodium fluoride | LD50 = 52 mg/kg ( Rat ) | > 2000 mg/kg (Rat) |                 |

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|                                            |                                                                                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| (b) skin corrosion/irritation;             | Category 2                                                                                                                    |
| (c) serious eye damage/irritation;         | Category 2                                                                                                                    |
| (d) respiratory or skin sensitization;     |                                                                                                                               |
| Respiratory                                | Based on available data, the classification criteria are not met                                                              |
| Skin                                       | Based on available data, the classification criteria are not met                                                              |
| (e) germ cell mutagenicity;                | Based on available data, the classification criteria are not met                                                              |
| (f) carcinogenicity;                       | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;                 | Based on available data, the classification criteria are not met                                                              |
| (h) STOT-single exposure;                  | Based on available data, the classification criteria are not met                                                              |
| (i) STOT-repeated exposure;                | Based on available data, the classification criteria are not met                                                              |
| Target Organs                              | None known.                                                                                                                   |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                                                       |
| Symptoms / effects, both acute and delayed | Other side effects may include nausea, headache, diarrhea, cramps, vomiting, or flu-like symptoms                             |

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component       | Freshwater Fish                                                                                                                                                                               | Water Flea                                              | Freshwater Algae                                                                                                              | Microtox |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------|
| Sodium fluoride | Lepomis macrochirus:<br>530 mg/L LC50 96 h<br>(static)<br>830 mg/L LC50 96 h<br>(semi-static)<br>Pimephales promelas:<br>180 mg/L LC50 96 h<br>Oncorhynchus mykiss:<br>38 - 68 mg/L LC50 96 h | 338 mg/L EC50 = 48 h<br>98 mg/L EC50 = 48 h<br>(static) | EC50: = 850 mg/L, 72h<br>static (Desmodesmus<br>subspicatus)<br>EC50: = 272 mg/L, 96h<br>(Pseudokirchneriella<br>subcapitata) | -        |

**Persistence and Degradability****Persistence****Degradability****Degradation in sewage treatment plant**

Soluble in water, Persistence is unlikely, based on information available.

Not relevant for inorganic substances.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the

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environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport****UN-No** UN1690  
**Proper Shipping Name** SODIUM FLUORIDE, SOLID  
**Hazard Class** 6.1  
**Packing Group** III**IMDG/IMO****UN-No** UN1690  
**Proper Shipping Name** SODIUM FLUORIDE, SOLID  
**Hazard Class** 6.1  
**Packing Group** III**IATA****UN-No** UN1690  
**Proper Shipping Name** SODIUM FLUORIDE, SOLID  
**Hazard Class** 6.1  
**Packing Group** III**Special Precautions for User** No special precautions required**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component       | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Sodium fluoride | X                                                   | X                                       | X    | X     | 231-667-8 | X    | X   | X     | X    | X    | X    | KE-31540 |

**National Regulations**

Sodium fluoride, Puratronic, 99.995% (metals basis excluding K), K 60ppm approx.

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 16-Nov-2010  
**Revision Date** 13-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**